INCH-POUND

MIL-C-50652/15B (AR)
16 May 1990
SUPERSEDING
MIL-C-50652/15A (AR)
30 May 1981

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548330
(CONVERTER, RANGE RATE D/A 4A17))

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 and applicable drawings comprise the complete requirements for the range rate D/A converter used in Range Computer, CP-888/VPS-2, part of Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-50652 - Circuit Card Assemblies General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548330 - Range Rate D/A Converter

PACKAGING DATA SHEET

PS10548330 - Packaging of Circuit Card Assembly, Range Rate D/A Converter

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548330, all associated drawings, and with all requirements specified in applicable specifications and standards. The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. As specified in paragraph 3.3 of MIL-C-50652.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram.

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3.5 Performance. For the requirements of 3.5.5 through 3.5.8 logic input levels shall be as follows:

Logic 0: 0.0 ±0.5 volts direct current (Vdc) level.
Logic 1: +2.8 to 5.4 Vdc level.

3.5.1 Input voltages. The input voltages shall be as specified in Table I. All voltages shall be regulated with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Bias voltage. The bias voltage measured at TP4 of the range rate D/A converter assembly shall be within the limits of +2.0 and +2.2 Vdc.

3.5.3 -310M/sec reference. The reference voltage measured at P1-12 and P1-14 shall be adjustable by R4 between the limits of -9.90 and -10.10 Vdc.

3.5.4 +310M/sec reference. The reference voltage measured at P1-16 shall be within the limits of +9.90 and +10.10 Vdc.

TABLE I. Input voltages

VOLTAGE	TERMINALS
+5.0 ±0.3 Vdc	P1-30, P1-31, P1-32
+5.0 V return	P1-24, P1-25, P1-26
+15.0 ±0.3 Vdc	P1-36, P1-37
±15.0 V return	P1-22, P1-23, P1-34, P1-35
-15.0 ±0.3 Vdc	P1-20, P1-21

3.5.5 Positive gain adjust. With the logic levels applied as specified in step 1 of Table II, the output voltage at P1-41 shall be adjustable by R15 between the limits as specified in step 1 of Table II.

3.5.6 Negative gain adjust. With the logic levels applied as specified in step 2 of Table II, the output voltage at P1-41 shall be adjustable by R16 between the limits as specified in step 2 of Table II.

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3.5.7 Range rate voltages. With the logic levels applied sequentially for steps 3 through 12 as specified in Table II the range rate output voltages at P1-41 shall be within the limits as specified in Table II.

3.5.8 Range rate ratios. With the logic levels applied sequentially as specified in Table III, the ratio of the range rate output voltage at P1-41 to the reference output voltage terminal specified in Table III, shall be within the limits as specified in Table III.

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TABLE II. Range rate voltages

STEP	EQUIVALENT1/ RANGE RATE METERS/SEC(M/S)	INPUT LOGIC LEVELS (REFERENCE P1 TERMINALS) 52	2	6	4	44	46	42	51	OUTPUT VOLTAGE (REFERENCE TERMINAL P1-41) NOT LESS THAN	NOT MORE THAN
1	+310	1	1	1	1	1	1	1	0	+9.900	+10.100
2	-310	0	0	0	0	0	0	0	1	-9.900	-10.100
3	+160	0	0	0	0	0	0	1	0	+4.990	+ 5.090
4	+ 80	0	0	0	0	0	1	0	0	+2.470	+ 2.570
5	+ 40	0	0	0	0	1	0	0	0	+1.210	+ 1.310
6	+ 20	0	0	0	1	0	0	0	0	+0.580	+ 0.680
7	+ 10	0	0	1	0	0	0	0	0	+0.265	+ 0.365
8	+ 5	0	1	0	0	0	0	0	0	+0.107	+ 0.207
9	+ 2.5	1	0	0	0	0	0	0	0	+0.028	+ 0.128
10	0	0	0	0	0	0	0	0	0	-0.050	+ 0.050
11	-160	0	0	0	0	0	0	1	1	-4.910	- 5.010
12	0	1	1	1	1	1	1	1	1	-0.050	+ 0.050

1/ Range rates are listed by their schematic value, a rounded value, except for the +310 and -310 which is the true maximum range rate for which this board is adjusted.

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TABLE III. Range rate ratios

STEP	EQUIVALENT RANGE RATE METERS/SEC (M/S)	INPUT LOGIC LEVELS (REFERENCE P1 TEMINALS) 52	2	6	4	44	46	42	51	RANGE RATE RATIO NOT LESS THAN	NOT MORE THAN	REF TERM P1	RATIO MODE
1	+310	1	1	1	1	1	1	1	0	+0.9802	+1.0202	16	+
2	-310	0	0	0	0	0	0	0	1	-0.9802	-1.0202	14	-
3	+160	0	0	0	0	0	0	1	0	+0.4940	+0.5141	16	+
4	+80	0	0	0	0	0	1	0	0	+0.2446	+0.2596	16	+
5	+40	0	0	0	0	1	0	0	0	+0.1198	+0.1323	16	+
6	+20	0	0	0	1	0	0	0	0	+0.0574	+0.0687	16	+
7	+10	0	0	1	0	0	0	0	0	+0.0263	+0.0369	16	+
8	+5	0	1	0	0	0	0	0	0	+0.0106	+0.0209	16	+
9	+2.5	1	0	0	0	0	0	0	0	+0.0028	+0.0129	16	+
10	0	0	0	0	0	0	0	0	0	-0.0050	+0.0050	16	+
11	-160	0	0	0	0	0	0	1	1	-0.5141	-0.4940	14	-
12	0	1	1	1	1	1	1	1	1	-0.0050	+0.0050	14	-

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3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Workmanship. The workmanship requirements of MIL-F-13926, MIL-S-45743 and MIL-S-46844 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. As specified in paragraph 4.1 of MIL-C-50652.

4.2 Classification of inspections. As specified in paragraph 4.2 of MIL-C-50652.

4.3 First article inspection. As specified in paragraph 4.3 of MIL-C-50652.

4.4 Quality conformance inspection. As specified in paragraph 4.4 of MIL-C-50652.

4.5 Special sampling. As specified in paragraph 4.5 of MIL-C-50652.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. As specified in paragraph 4.6.1 of MIL-C-50652.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. As specified in paragraph 4.6.2.1 of MIL-C-50652.

4.6.2.2 Contractor design. As specified in paragraph 4.6.2.2 of MIL-C-50652.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE IV. Test equipment list

ITEM NUMBER		DESCRIPTION
1.	Power supply	5.0 ±0.3 Vdc, regulated; ac ripple 50 mV peak-to-peak maximum; capacity, 200 milliamps (mA).
2.	Power supply	15.0 ±0.3 Vdc, regulated; ac ripple 50 mV peak-to-peak maximum; capacity, 200 mA
3.	Power supply	Same as item 2.
4.	Digital volt-ratio meter	Visual display: 5 digits Dc range: 0.0000 to ±999.99 V. Accuracy ±10.01% of reading +0.0001% of full scale. Dc input resistance: Greater than 10 megohms. Dc ratio: 1.0999 to 1, ±0.005% of reading +0.002% of full scale with ±10 V minimum reference. Common mode rejection: Dc greater than 134 dB, 60 Hz greater than 120 dB. Response time: 1500 milliseconds (ms) maximum.
5.	Test fixture	As specified in Figure 1. See Table V for test fixture parts list.

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TABLE V. Test fixture parts list

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1 thru E4	Binding post: H.H. Smith 257 or equivalent.
J1	Connector: any of the following or equivalent.
	(1) 5893232, Malco Mfg. Co., Inc., Chicago, Illinois
	(2) 500946-01, National Connector Corp. Minneapolis, Minn.
	(3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
S1 thru S8	Switch SPDT
TP1 thru TP5	Tip jack: H.H. Smith 1501 or equivalent

4.7 Methods of inspection.

4.7.1 Environmental. As specified in paragraph 4.7.1 of MIL-C-50652.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. The range rate D/A converter assembly shall be tested as follows: Connect range rate D/A converter and test equipment as shown in Figure 2. Switches S1 through S8, on the test fixture (Table IV, item 7) are used to supply the logic level input signals as specified in Tables II and III. These switches supply a logic 1 in position 1 and a logic 0 in position 2.

4.7.3.1 Input voltages. Using power supplies (Table IV, items 1 through 3), apply input voltages as specified in Table I.

4.7.3.2 Bias voltage. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the bias voltage TP4 of the range rate D/A converter to be as specified in 3.5.2. Use TP5 on the test fixture (Table IV, item 5) as ground reference.

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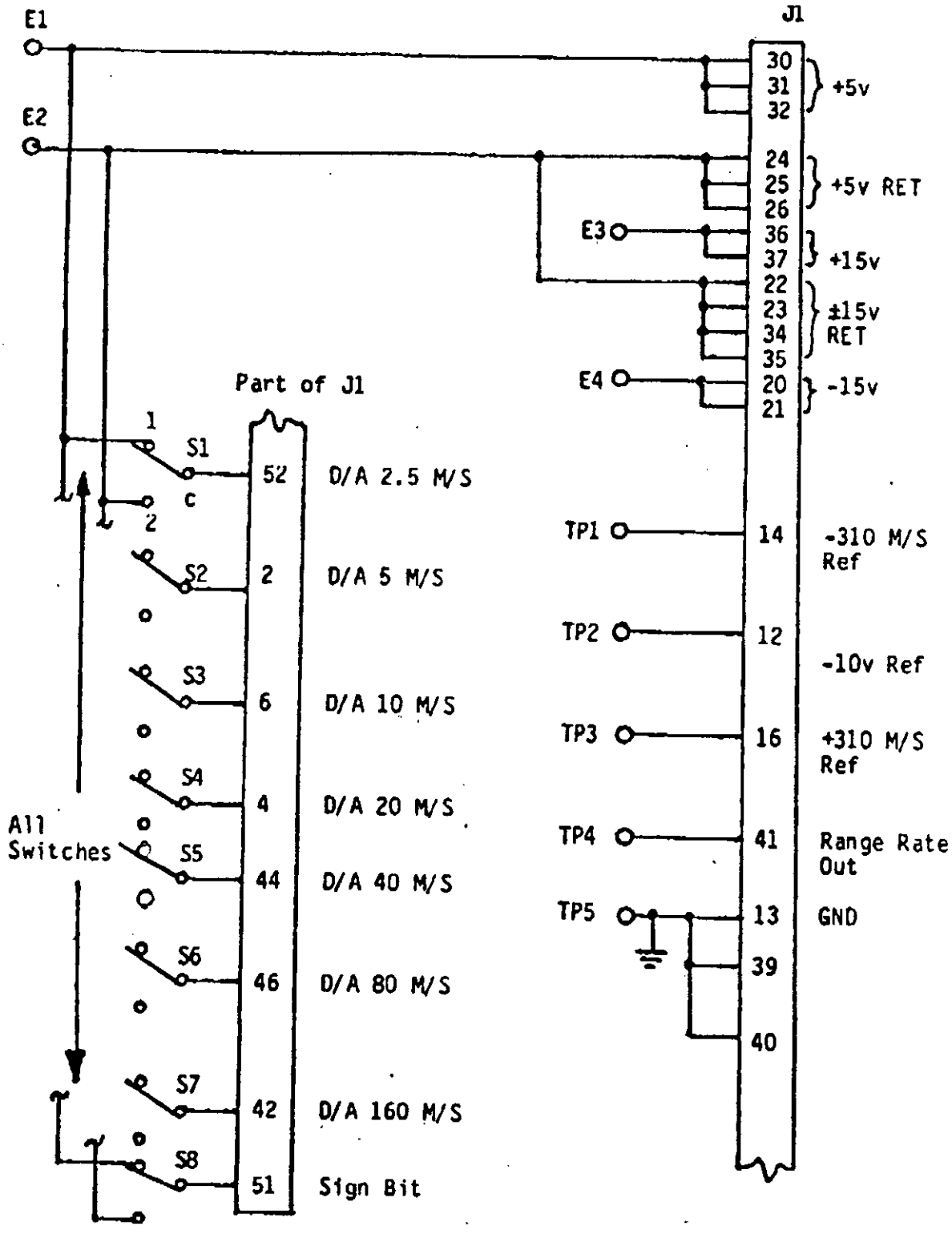


FIGURE 1. Test Fixture setup.

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4.7.3.3 -310 M/sec reference. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the -310 volt reference at TP1 and TP2 on the test fixture to be as specified in 3.5.3. Use TP5 on the test fixture (Table IV, item 5) as ground reference.

4.7.3.4 +310 M/sec reference. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the +310 volt reference at TP3 on the test fixture to be as specified in 3.5.4. Use TP5 on the test fixture (Table IV, item 5) as ground reference.

4.7.3.5 Positive gain adjust. Using switches S1 and S8 on the test fixture (Table IV, item 5) apply the logic levels as specified in step 1 of Table II. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the positive gain adjust at TP4 on the test fixture to be as specified in 3.5.5. Use TP5 on the test fixture as ground reference.

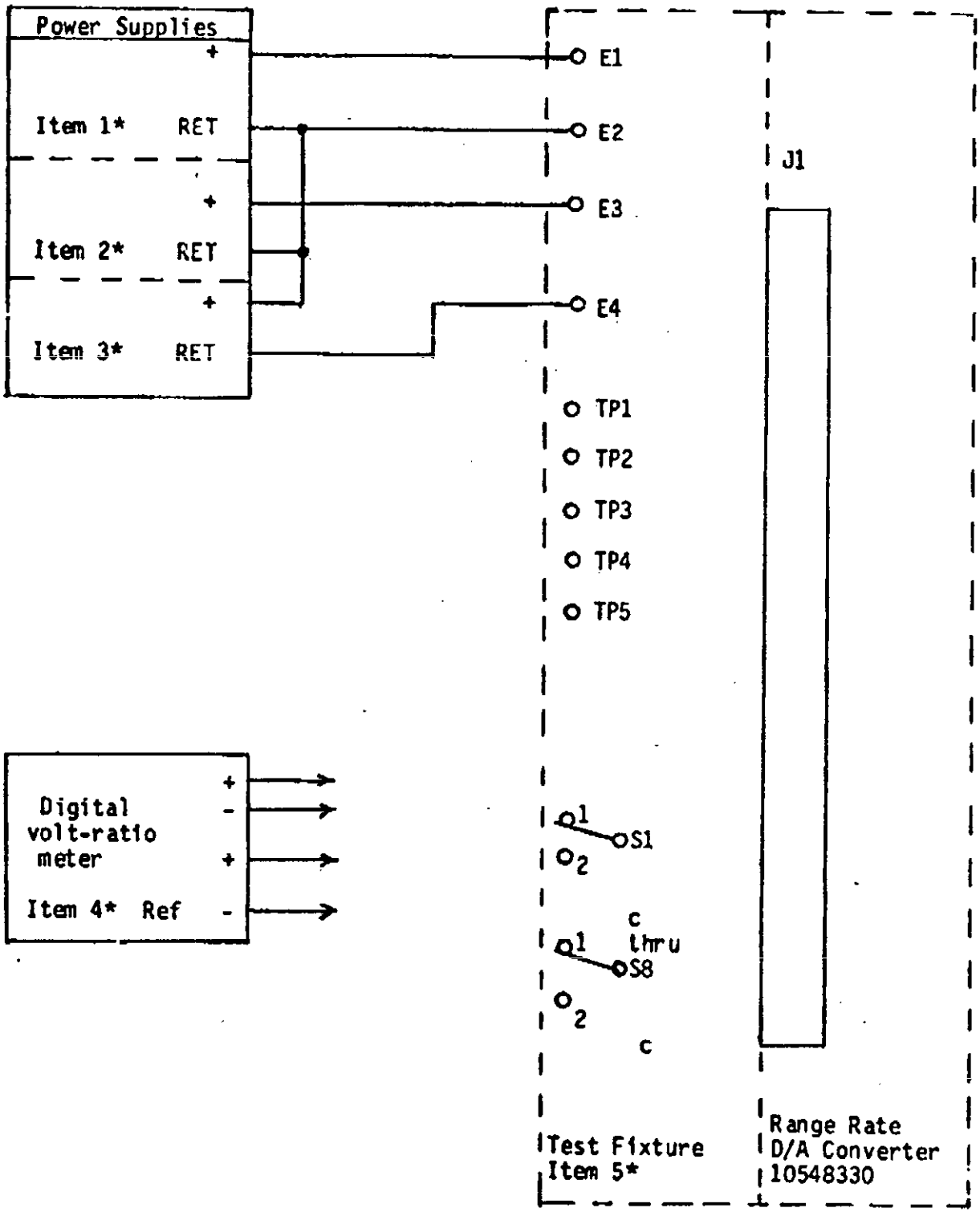
4.7.3.6 Negative gain adjust. Using switches S1 through S8 on the test fixture (Table IV, item 5) apply the logic levels as specified in step 2 of Table II. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the negative gain adjust at TP4 on the test fixture to be as specified in 3.5.6. Use TP5 on the test fixture as ground reference.

4.7.3.7 Range rate voltages. Using switches S1 through S8 on the test fixture (Table IV, item 5) sequentially apply the logic levels as specified in steps 3 through 12 of Table II. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the range voltages at TP4 on the test fixture to be as specified in 3.5.7. Use TP5 on the test fixture as ground reference.

4.7.3.8 Range rate ratios. Using switches S1 through S8 on the test fixture (Table IV, item 5) sequentially apply the logic levels as specified in the steps of Table III. Using the volt-ratio meter (Table IV, item 4) in the ratio mode, measure the range rate ratios to be as specified in 3.5.8. Connect the volt-ratio meter leads to the test fixture test points as follows:

+ voltage lead	- TP4
- voltage lead	- TP5
+ reference lead	- TP1 for P1-14 (- ratio mode) TP3 for P1-16 (+ ratio mode)
- reference lead	- TP5

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*Refer to 4.6.2.3

FIGURE 2. Range rate D/A converter test setup.

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5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS10548330. The level of protection shall be as specified in the procurement document.

6. NOTES. As specified in paragraph 6 of MIL-C-50652.

Custodian:
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	MIL-C-50652/15B (AR)	16 May 1990

3. DOCUMENT TITLE

CIRCUIT CARD ASSEMBLY: 10548330 (CONVERTER, RANGE RATE D/A 4A17)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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